

Experiment 4: Fixed Height + Obstacle + Multi Wind Strength | strong_wind | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1830	6.832	5.094	15.082	5.2%	20.4%	35.8%	20.078	47.338	13.451	41.523	0.680	1.425	1.522
PPO-GRU	1731	5.810	2.502	16.511	22.0%	45.5%	53.3%	21.142	59.346	11.507	28.412	0.596	1.383	1.539
PPO-RNN	1704	6.017	2.997	16.457	14.6%	35.6%	50.1%	23.724	71.459	10.923	26.653	0.657	1.466	1.601
MPC	437	6.857	2.659	22.123	11.4%	31.8%	59.3%	2.805	4.266	1.811	3.700	0.155	0.181	0.271

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1830	91.5%	170	8.5%	8.2%	165	97.1%	5	0
PPO-GRU	2000	1731	86.5%	269	13.4%	10.4%	209	77.7%	60	0
PPO-RNN	2000	1704	85.2%	296	14.8%	13.0%	259	87.5%	37	0
MPC	2000	437	21.9%	1563	78.2%	78.2%	1563	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	1563	1408	90.1%	990	70.3%	7.339
PPO-GRU	1563	1298	83.0%	642	49.5%	6.131
PPO-RNN	1563	1267	81.1%	655	51.7%	6.197